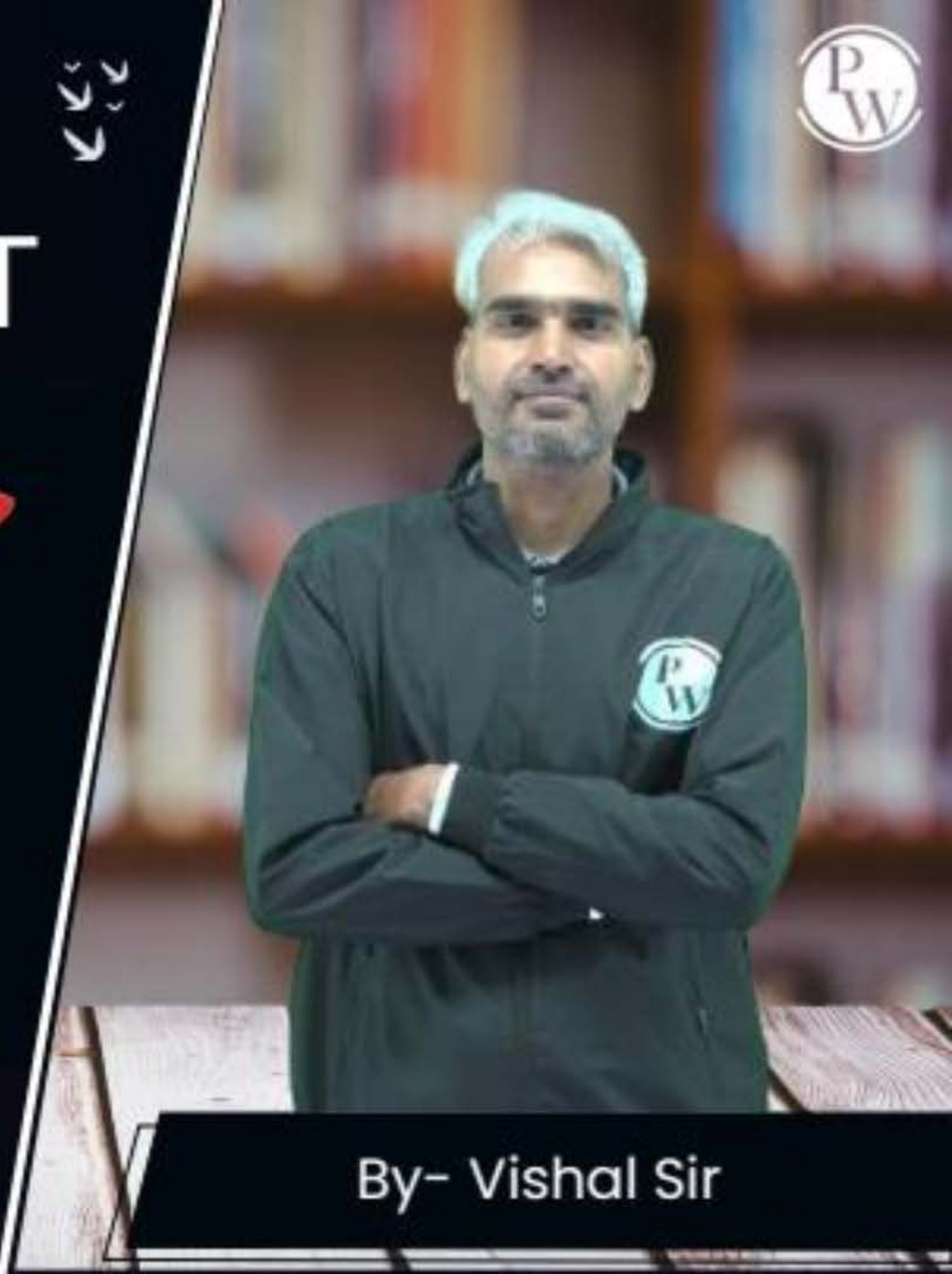
Computer Science & IT



Database Management System

Query Languages

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



Topic Practice questions of Relational Algebra



Slide

Topics to be Covered



- Topic SQL
- Topic SQL clauses
- Topic Aggregate functions

Slide



Topic : SQL

Prerequisite :- Relational Algebra



→ Structured Query language

→ It is a non-procedural query language

∴ we must understand the "Syntax" of SQL



Topic : Commands in SQL

Commands in SQL

- Data definition Language (DDL)
  - Create
  - Drop
  - Alter
- Data Manipulation language (DML)
  - Insert
  - Delete
  - Update
  - Truncate
- Data Control language (DCL)
  - Grant
  - Revoke
- Transaction Control Language (TCL)
  - Commit
  - Rollback/Abort
- Data query language (DQL)
  - Select



Topic : SQL clauses

1. From
2. Where
3. Group by
4. Having
5. Order by

Basic Syntax of SQL query :

It is optional
If we do not use "distinct" then SQL query output may contain duplicate tuples

$\pi_{A_1, A_2, \ldots A_K}$ ≡ ① Select distinct $A_1, A_2, \ldots, A_K$

List of attributes required in output

$R_1 \times R_2 \times \ldots \times R_m$ ≡ ② From $R_1, R_2, \ldots, R_m$

List of database tables required for the execution

$\sigma_{(\text{Cond}^n \text{ to select tuples})}$ ≡ ③ Where (Condition to select tuples)

No Equivalent R.A. Expression

④ Group by (Attributes)

⑤ Having (Cond^n to select groups)

⑥ Order by (Attributes) DESC / ASC

Projection (π) in R.A. $\not\equiv$ Select in SQL

because duplicate may be present in SQL

Projection (π) in R.A. $\equiv$ Select distinct in SQL



Topic : Aggregate functions

There are five different aggregate functions in SQL

1. Count ( )
2. Sum ( )
3. Avg ( )
4. Max ( )
5. Min ( )



Topic : Aggregate functions

- Select Count (*) From STUDENT=
- Select Count (Sid) From STUDENT =
- Select Count (distinct Sid) from STUDENT=
- Select Count (Marks) From STUDENT =
- Select Count (distinct Marks) From STUDENT=

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

Everything

- Count(*) - It returns the total number of tuples in the relation. Count(*) will also count the empty tuples, i.e., the tuples in which all values all NULL.

- Count(Attribute) - It will count the total number of non NULL values in the column corresponding to the attribute specified with count function.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Select Count (*) From STUDENT = 7
- Select Count (Sid) From STUDENT = 6
- Select Count (distinct Sid) from STUDENT = 6
- Select Count (Marks) From STUDENT = 5
- Select Count (distinct Marks) From STUDENT = 4

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : NOTE

- NULL is a non-zero unknown value.
- No two NULL are equal. {∵ Comparison operators are not used to compare two NULL values}
- NULL values are always discarded by aggregate functions except for count(*).
- Arithmetic operation with NULL will always produce NULL

NULL + 50 = NULL



Topic : Aggregate functions

- Sum(Attribute) - It will return the summation of all non NULL values in the column corresponding to the attribute specified with aggregate function Sum().

- Sum(distinct Attribute) - It will return the summation of all distinct non NULL values in the column corresponding to the attribute specified with aggregate function Sum().

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Select Sum (Marks) From STUDENT = 250

- Select Sum (distinct Marks) From STUDENT = 190

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Avg(Attribute) - It will return the average of all non NULL values with respect to specified attribute.

$$\text{Avg (Attribute)} = \frac{\text{Sum (Attribute)}}{\text{Count (Attribute)}}$$

- Avg(distinct Attribute) - It will return average of all distinct non NULL values with respect to specified attribute.

$$\text{Avg (distinct attribute)} = \frac{\text{Sum (distinct attribute)}}{\text{Count (distinct attribute)}}$$

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Select Avg (Marks) From STUDENT = 50

$$= \frac{\text{Sum (Marks)}}{\text{Count (Marks)}} = \frac{250}{5} = 50$$

- Select Avg (distinct Marks) From STUDENT:

$$= \frac{\text{Sum (distinct Marks)}}{\text{Count (distinct Marks)}} = \frac{190}{4} = 47.5$$

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Max(Attribute) - It will return the maximum of all the values in the column corresponding to the specified attribute.

- Min(Attribute) - It will return the minimum of all the values in the column corresponding to the specified attribute.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Select Max (Marks) From STUDENT = 70

- Select Min (Marks) From STUDENT = 20

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



2 mins Summary

Topic SQL

Topic SQL clauses

Topic Aggregate functions

* Home Work :-

- Read about following SQL Clauses.

1. From
2. Where
3. Group by
4. Having

H.W. Try to create a table using Create Command.

Slide

THANK - YOU